

Introducing Fractals

We will now put aside chaos and dynamical systems and focus instead on a different topic: fractals. This will not be a complete departure from what has come before; we will see that iteration plays a key role in the generation of fractals. Gradually over the next several chapters we will see that fractals and the sorts dynamical systems we have studied in the first two parts of this book are closely related.

15.1 Shapes



Figure 15.1 shows three familiar shapes from geometry: a circle, a line segment, and a rectangle. These geometric forms are abstractions of shapes that we encounter in the physical world: a round coffee mug, a clothesline, and a tabletop. But the world we live in is much richer than this. There are many shapes—the branches of a tree, the bumps of a mountain range, the meander of a river—that do not resemble the shapes of Fig 15.1. Consider the images shown in Fig. 15.2. These objects from the natural and physical world are very different from the simple circles and lines of ordinary geometry. Sure, we could describe a winding river or branching trees as a collection of line segments arranged in a particular way. But it seems that this would be missing the essence of the shape that we are trying to describe.

The desire to better describe forms such as trees and winding rivers leads us to a different sort of shape. We construct an initial example by an iterative process, shown in Fig. 15.3. At $n = 0$ we start with a small square. We can think of this as a seed, playing the role that the initial condition x_0 does for an iterated function. To get to the shape at step $n = 1$, we make four copies of the shape at $n = 0$, and place one copy at each of the corners. We then repeat, or iterate, this process. To get to step $n + 1$ we take the shape at n , make four copies of it, and place one copy at each of the corners. The result, as one applies this rule over and over and over, is an intricate (and very large) structure that resembles a snowflake. This snowflake shape is known as a **fractal**. I will describe

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Fig. 15.1 Three shapes from ordinary geometry: a circle, a line segment, and a rectangle.

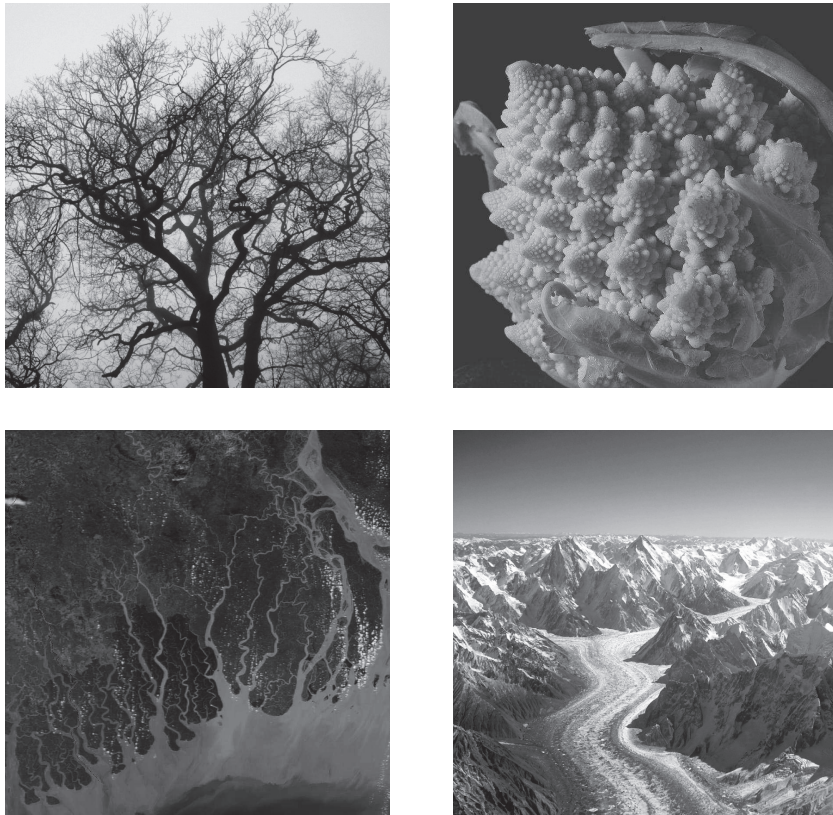


Fig. 15.2 Four naturally occurring objects that are well viewed as fractals. Clockwise from top left: tree silhouette (Bruce Thompson, licensed under Creative Commons CC BY-NC-SA 2.0); romanesco broccoli (Licensed under Creative Commons CC0, photo courtesy PDPPhoto.org); Baltoro glacier and the Karakoram mountains (Guilhem Vellut, licensed under Creative Commons Attribution-Share Alike 2.0 Generic license); and the Ganges river delta (NASA Earth Observatory).

fractals more fully below, but for now the main thing to note is how different it is than a simple square or circle.

For our next example of a fractal, consider Fig. 15.4. Again, this fractal is constructed via iteration. We start at step $n = 0$ with a line segment. We then remove the middle third of that line segment to obtain the shape labeled $n = 1$. Repeating this step, we get the shape at $n = 2$. That is, we remove the middle third of each line segment at $n = 1$. We keep on doing this, removing at every step the middle third of every line segment. If we carry this process on forever, the result is an infinite number of very, very, very tiny line segments.¹ The collection of these minuscule line segments is known as the Cantor set.

¹Strictly speaking, in the limit that n goes to infinity, the tiny line segments approach points.

15.2 Self-Similarity

The Cantor set and the snowflake fractal are both self-similar. What this means is that a small portion of them looks like the whole. For

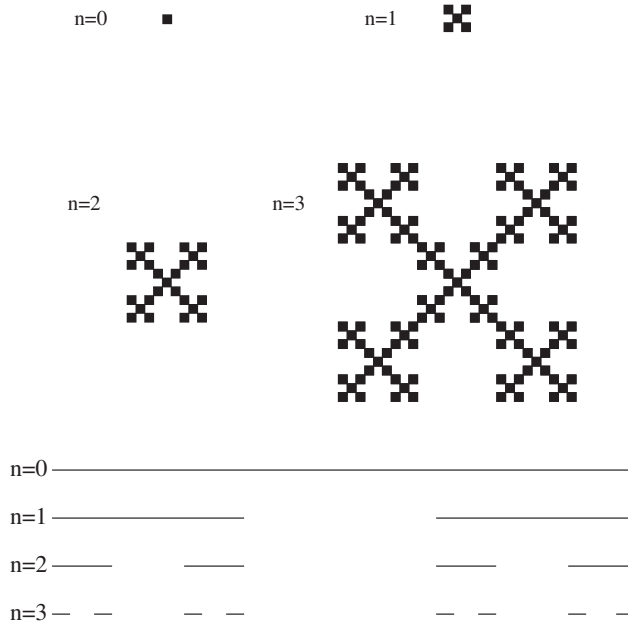


Fig. 15.3 The first several stages in the construction of a “snowflake” fractal. One starts at $n = 0$ with a seed shape, in this case a small square. One builds the next shape by placing a copy of the previous shape on each of the four corners of the current shape.

Fig. 15.4 The first several stages in the construction of the middle-thirds Cantor set. One starts with a line segment. At every successive step, one removes the middle third of every line segment in the previous step.

example, imagine breaking off an arm of the snowflake. What you have looks like a miniature copy of the entire snowflake. You could magnify the small portion and you will get a shape that looks like the whole snowflake. You could break a smaller arm off of the arm you already broke off, and again you would have something that is a miniature copy of the whole snowflake. The same property is true of the Cantor set, although this might be a little bit harder to see. If you take a small portion of the set you will have something that is a miniature copy of the entire set.

The natural fractals shown in Fig. 15.2 are also self-similar. Here the self-similarity is not exact; small portions of each picture are not identical to the large picture, but rather bear a close resemblance. For example, a branch of a tree looks similar, but not identical, to the full tree.

This property is called **self-similarity**; an object is self-similar if it contains replicas of itself of many different sizes. Fractals are self-similar geometric objects. As a counter-example, consider a person. People are not fractals. If you break off a person’s arm, what you have in your hands will look like an arm, and not a small copy of the person. So a person is not self-similar, and hence is not a fractal.

Note that there are some non-fractals that are nevertheless self-similar. An example is a line segment. If you break off a portion of a line segment, you have another line segment, which looks just like the original line segment. However, this is not considered a fractal, for reasons that will be discussed in Chapter 16. For now, we can think of fractals as being geometric shapes that are self-similar, but in a “non-trivial” or “non-boring” way. Line segments are self-similar in a boring fashion, and

hence are not considered fractals.

Here is another way to think about self-similarity. Suppose one day you woke up and you had significantly changed size. Perhaps you wake up Monday morning and you are only six inches tall. How could you tell? A clear indication would be that you were suddenly smaller than your pillow, your alarm clock would appear gigantic, it might not be safe to jump out of bed, and your cat might eat you. In brief, it would be immediately and stunningly obvious to you that you were very small.

However, suppose you lived in a fractal world—perhaps in an arm of the fractal snowflake of Fig. 15.3. Remember that this fractal gets very, very large as n , the number of times we apply the snowflake-building rule, gets large. If you lived on this snowflake, you would not be able to tell that that you had been shrunk. There are no clues as to scale or size. The entire universe consists of snowflakes within snowflakes within snowflakes. So there is no way to tell how big you are. The only thing you can measure yourself with is a snowflake, and there are snowflakes of all different sizes, so you can never really tell how big you are.

15.3 Typical Size?

As discussed in the preceding paragraphs, for a fractal there is not a typical size. For the fractal snowflake, the snowflake motif is repeated again and again and again at different sizes. So it does not really make sense to talk about an average snowflake size. The Cantor set of Fig. 15.4, after we have iterated the construction process many times, consists of points that are clustered together. But these clusters are clustered into clusters, which in turn are clustered into bigger clusters, and so on.

For ordinary, non-fractal objects we are used to describing them by stating their size: a circle with a radius of two inches, a chair that is 3.5 feet tall, a track on a CD that lasts five minutes. For more complicated situations, we often resort to averages or statements about the size of a typical instance of some entity. For example, we might say that a typical person is 5.5 feet tall, an average cat weighs 9 pounds, and most pints of ice cream cost around \$2.50.

But this type of description does not work for fractals; stating the average size of a cluster in the Cantor set or the size of a typical snowflake motif in the snowflake fractal does not really capture the essence of the shape. In fact, we shall see in Chapter 19 that there are some fractals for which the concept of an average size is not well defined. But this begs the question: how *can* we describe fractals if not with averages? The answer to this question will come in the next chapter, where we will see that the geometric notion of dimension can be extended so as to capture some of the structural features of fractals.

15.4 Mathematical vs. Real Fractals

Finally, a few words about mathematical versus real fractals. The fractal snowflake and the Cantor set are abstractions. Strictly speaking they are defined by the result of an infinite iteration process. As such, they have structure on all scales. For example, you could keep zooming in on such a mathematical fractal and you would see the same shape repeated endlessly.

The same is not true for real fractals. Consider, for example, a fern. We can zoom in on the fern, and we will see smaller copies of the fern. But eventually this stops, and the fern no longer looks like little ferns. Instead we start to see individual cells. Or, consider the winding path of a river. We could imagine starting with an aerial view taken by a satellite, and then start zooming in. For quite some time we could see a self-similar shape; meanders on top of meanders, wiggles on top of wiggles. But eventually we will zoom in to the river itself, and we would just see water. Similarly, if we started with the satellite image and zoomed out, eventually the river would disappear from view. So real fractals have some cut-off sizes above and below which an otherwise self-similar object fails to appear self-similar. In contrast, mathematical fractals have no such cut-off; in principle we could keep zooming in forever, and see ever-smaller copies of our original shape.

This does not mean, however, that mathematical fractals are useless for describing the real world. It is important to remember that a fractal, like any geometric object, is just an abstraction or an idealization. For example, consider a perfect, mathematical circle. Such a thing is just an idea, albeit a very useful one. In Fig. 15.1 I have shown a drawing of a circle, made with the graphics program `xfig`. It is a nice circle, but it is not perfect. Look closely enough—you might need a magnifying glass—and you will see imperfections. And in nature, in the physical world in which we live, there certainly are no perfect circles, just shapes that approximate circles to varying degrees. Nevertheless, it would be hard to argue against the geometric idea, or ideal, of a circle. It certainly is a useful approximation or abstraction of things we encounter in the physical world.

So it is with fractals. In the real world there are no perfect fractals. But many have found that fractals are a tremendously useful and evocative way of capturing the qualities of shapes and processes that are, to varying degrees, self-similar.

Exercises

- (15.1) Produce a real-life example of an object or phenomena that is a fractal. Explain why your example fits the criteria for being a fractal.
- (15.2) The first two steps ($n = 0$ and $n = 1$) in the construction of a fractal known as the Sierpiński triangle are shown in Fig. 15.5. Sketch the shape for $n = 2$ and $n = 3$.

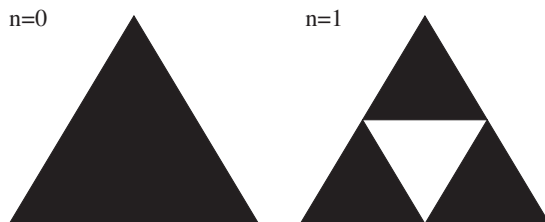


Fig. 15.5 The first two steps in the construction of the Sierpiński triangle.

- (15.3) The first several rows of Pascal's triangle are shown in Fig. 15.6. Pascal's triangle is constructed as follows. The top and edges of the triangle are filled with 1's. Other entries in the triangle are obtained by adding the two entries directly above it. For example, the hexagons with the number 4 in them in Fig. 15.6 both have a 1 and a 3 above them.

- (a) Continue the triangle so that it extends for fifteen to twenty rows. Use hexagonal graph paper, which you can download at <http://incompetech.com/graphpaper/hexagonal/>.
- (b) Now shade in all the cells that have an odd number in them.
- (c) Does the resultant shape look familiar?

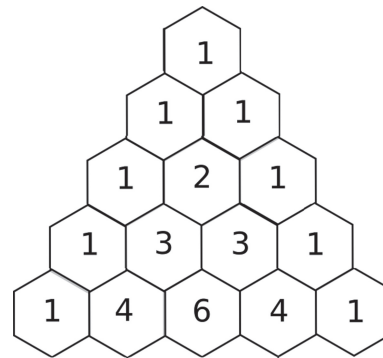


Fig. 15.6 The beginning of Pascal's triangle. The value in each hexagon is equal to the sum of the two hexagons above.